Project Title: Import Data Using Transform Map – Spreadsheet

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Team ID: SWTID-2026-2519

PHASE 3 – PROJECT DESIGN

1. Introduction:

The Project Design phase describes the structure and workflow of the proposed ServiceNow spreadsheet data import system.

The design focuses on importing spreadsheet records into the Institution Details table using an Import Set, Import Set Table, Transform Map, field mappings, and matching logic.

2. System Architecture:

The project follows the following data-import structure:

Spreadsheet → Import Set → Import Set Table → Transform Map → Field Mapping → Target Table

The spreadsheet provides the source data. ServiceNow stores the uploaded records in the Import Set Table. The Transform Map processes the source records and maps them to the corresponding fields in the Institution Details target table.

3. System Workflow:

The basic workflow is:

1. User prepares the spreadsheet with required data.
2. User uploads the spreadsheet into ServiceNow.
3. ServiceNow creates an Import Set and stores the imported rows.
4. The Import Set Table holds the source data temporarily.
5. ServiceNow checks the configured Transform Map.
6. Source fields are mapped to the target fields.
7. Matching logic checks the Student Roll Number.
8. Records are created or updated in the Institution Details table.
9. Imported records are verified in the target table.

Sample Workflow Diagram:

Spreadsheet → Upload → Import Set → Transform Map → Field Mapping → Institution Details

4. Table Design:

The custom target table used in the project is:

Label: Institution Details

Table Name: u_institution_details

Field	Field Type	Purpose
Student Roll Number	Auto Number	Unique student identification / matching
Student Name	Reference – User	Stores student information
Faculty Name	Reference – User	Stores faculty information
Branch	Choice	Stores EEE, ECE, or CSE
Email	String	Stores email address
Phone Number	String	Stores phone number
Description	Multi String	Stores additional information

5. Spreadsheet Source Design:

The spreadsheet is designed as the source file for the import process. The available columns are:

• Student Roll Number

• Student Name

• Faculty Name

• Branch

• Email

• Phone Number

• Description

The Branch values may be EEE, ECE, or CSE.

6. Import Set Design:

The Import Set is designed to receive the spreadsheet data before transformation. An Import Set Table stores the uploaded source records temporarily.

Import Set Components:

■ Source: Spreadsheet

■ Import Set Table: Temporary source table

■ Target Table: Institution Details

■ Import Status: Ready for transformation

7. Transform Map Design:

A Transform Map is designed to connect the Import Set Table with the Institution Details target table.

Transform Map Component	Design

Source Table	Import Set Table
Target Table	u_institution_details
Active	Yes
Run Transform	Enabled
Matching Field	Student Roll Number

8. Field Mapping Design:

The source spreadsheet fields are mapped to the corresponding target fields as follows:

Source Field	Target Field
Student Roll Number	Student Roll Number
Student Name	Student Name
Faculty Name	Faculty Name
Branch	Branch
Email	Email
Phone Number	Phone Number
Description	Description

9. Transform Logic:

The Transform Map processes each Import Set record and transfers the source values into the matching fields of the Institution Details table. Student Roll Number is used as the matching field so that existing records can be identified during transformation.

10. Data Import Matrix:

Source Data	Transform Action	Target Result
Student Roll Number	Match / Map	Student Roll Number
Student Name	Map	Student Name
Faculty Name	Map	Faculty Name
Branch	Map	Branch
Email	Map	Email
Phone Number	Map	Phone Number
Description	Map	Description

11. Security Design:

The system is designed to protect the import process using controlled ServiceNow access and proper data validation.

■ User authentication

■ Authorized import access

■ Controlled Transform Map configuration

■ Field mapping validation

■ Matching-field validation

■ Target-table permissions

This design ensures that spreadsheet data is processed through the configured ServiceNow import workflow.

12. Expected Design Outcome:

The final design should provide:

■ Structured spreadsheet data import.

■ Import Set based source-data handling.

■ Transform Map based data transformation.

■ Correct source-to-target field mapping.

■ Matching of existing records using Student Roll Number.

■ Creation or updating of Institution Details records.

■ A structured and maintainable import configuration.

13. Conclusion:

The Project Design phase defines the architecture, spreadsheet source structure, Import Set, Transform Map, field mappings, workflow, matching logic, and security design of the system. This design will be used as the blueprint for implementing the Import Data Using Transform Map – Spreadsheet project in ServiceNow.